Digital transformation of telecom providers management customer system: a process research and effects assessment

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ABSTRACT

The article discusses the impact of the digital transformation of loyalty systems on the success rates of the telecommunications business. The specifics of the telecommunications industry are particularly high rates of customer churn, averaging from 10 to 67% per year. As a result, the cost of attracting a new client is 4-10 times more expensive than retaining an existing one. So digital transformation of customer loyalty systems becomes a tool for predicting customer behavior and delivering timely and relevant offers, improving the user experience. The purpose of the study is to propose the approach to managing of the digital transformation of customer loyalty systems of telecommunications companies by the measures of the business process management improving based on e-TOM model. The paper defines criteria for evaluating the effectiveness of customer loyalty systems digital transformation for telecommunications companies. The essence of the assessment method is to compare two blocks of indicators: economic efficiency coefficients (before and after digital transformation) and a group of indicators of effects accompanying digital transformation (indicators of strategic, client, operational, cultural, technological effects). The information base of this study was modern literature and statistical about digital transformation of telecommunications companies, as well as data of business processes of telecommunications sector organizations (from America, Europe, Asia, Russia) and the results of interviews with industry experts. Also, we used data from the report of the customer loyalty system's digital transformation results of the telecommunications company, conducted with the participation of the authors. The results of assess the effectiveness of the digital transformation of the telecom provider's customer loyalty system show an increase in the efficiency coefficient from 0.62 to 1.08, as well as an increase in all groups of effects except of personnel culture in the company. In particular, the customer satisfaction index (CSI) increases from 21% to 79% and the outflow of customers decrease from 37.5% to 25%. Finally, there was a discussion of the results of our study, and the conclusions were summarized.

CCS CONCEPTS

• **Applied computing** → Law, social and behavioral sciences; Economics.

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KEYWORDS

Digital transformation, customer loyalty system, telecommunication companies, telecom provider, eTOM model, effectiveness business processes

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1 INTRODUCTION

Digitalization is a global trend at the current stage of the information society development and the digital economy formation in various countries. The theoretical background of this research is the resource-based theory The expansion of the Internet to all spheres of life, the use of big data and artificial intelligence technologies for the purposes of business development leads to a total digital transformation of all business processes and activities [1–3] including the risk management [4]. The use of modern technologies such as artificial intelligence in order to predict user behavior and meet user needs accompanies the transition to Industry 4.0 [5–7].

Digital transformation for every company is a complex process that requires the implementation of operational and management changes focused on satisfying customer needs and User-Centric approach [8]. At the same time, in order to ensure the possibility of managing the digital transformation of the business, it is necessary to assess the effects of the digital transformation implementation and to plan the key performance indicators in advance [9]. However, the digital transformation is deep process touched every company's business processes so special attention should rather be paid not to the technologies and digital tools themselves [10–12], but to changes in organizational culture and business processes management [13–15].

The processes of modern technologies implementation in the telecommunications sector are occurring constantly and rapidly, which is a special characteristic of this high-tech business. There are about 7.9 billion active mobile subscribers worldwide and over 1.1 billion fixed broadband subscribers. At the same time, due to the potential for deployment of 5G networks, an increase in mobile technologies trend is expected [16]. The dynamics of the data transmitted around the world demonstrates an annual increase of about 20-25% exabytes of transmitted data [17].

The Russian telecommunications market is showing positive dynamics and the volume about $ 25 million in 2020 [18]. The number of broadband Internet users has grown by 2.1% (to 36.1 million people) and a steady growth of the market at 2% per year has expected [19]. At the same time, there is a scenario of a decrease in the growth dynamics progressively as the possibilities for increasing the average profitability per subscriber, coming from communication services provision, are exhausted. So, the telecommunications market is witnessing a process of commodization and downscaling of telecommunications operators' business.

The further development of telecommunications companies depends on their ability to build loyal relationships with users. With the digital transformation of business, opportunities arise for using modern technologies in the marketing of relationships, with customers, where the main indicator of success is customer loyalty and customer retention. The profitability of the business directly depends on these characteristics. After all, the members of customer loyalty programs usually spend up to 18% more than other customers [20]. And as little as 8% of loyal customers provide digital business with up to 41% of its income [21]. This is why six out of ten marketing teams monitor customer satisfaction and retention [22].

Loyalty systems are already an established feature of the retail market in such countries as the United States and Great Britain, European and Asian countries [23]. There is a positive correlation between the availability of a customer loyalty program and the competitiveness of supermarkets [19].

However, the telecommunications industry specificity consists in particularly high rates of customer outflow, averaging from 10 to 67% per year [24]. At the same time, the level of customer loyalty among telecommunications companies remains low: about 75% of the 17-20 million subscribers who sign an agreement with a new operator every year come from another communication provider. As a result, attracting a new client costs a telecommunications business 4-10 times more than retaining an existing one [17, 25].

The introduction of modern technologies such as artificial intelligence and big data into customer loyalty systems is becoming a tool for predicting customer behavior [26] and delivering timely and relevant offers improving user experience.

The importance of managing the digital transformation of loyalty systems in telecommunications companies is undeniable. However, there are several problems: on the one hand, the existing literature does not pay enough attention to improving customer loyalty systems of a telecommunications company, and on the other hand, from the point of view of managing business processes of telecommunications companies, there is an equal lack of attention to loyalty systems in the internationally accepted e-TOM system (Business Process Framework).

Therefore, the aim of the paper is to propose an approach to digital transformation management of the customer loyalty systems for telecommunications companies. It is needed to analyses of the business process management standard e-TOM and development of the method of assessing the effectiveness of the digital transformation management of the customer loyalty systems for telecommunications companies.

2 MATERIALS AND METHODS

This study is a literature and statistical review of works on business processes of organizations in the telecommunications sector (from America, Europe, Asia and Russia) and also interview of the experts in the digital transformation management of the telecommunications business, such as Netcraker. Also, we used data from the report of the customer loyalty system's digital transformation results of the Albania's telecommunications company, conducted with the participation of the authors. But due to the existence of a trade secret we cannot disclose the name of the company or use full data of that report.

The issues of the impact of the digital transformation of loyalty systems on business success indicators are examined. An approach to the digital transformation management of the customer loyalty system of telecommunication companies, based on eTOM model is proposed [27].

As the managing process the digital transformation of the customer loyalty system consists of the following stages:

- Stage 1 – company analysis. The digital maturity of the company and the processes of building loyal relationships with customers are assessed. A company analyzes such indicators as the nature of building relationships, the index of customer satisfaction with the program and the loyalty system, the use of technologies for the selection of customer-oriented offers, the method of access to the loyalty system and the amount of customer churn. The conclusion is made about the need for digital transformation.
- Stage 2 – preparation for the digital transformation. A strategy for the digital transformation of a specific company and a roadmap are created, and performance indicators are developed. The system and the loyalty program are developed, as well as the training of human resources for its implementation with orientation at the eTOM model.
- Stage 3 – pilot implementation support. At the stage of the pilot implementation of technologies, testing and monitoring of loyalty systems takes place on a test network of customers with technical and customer support.
- Stage 4 – assessment of intermediate indicators of the effectiveness of digital transformation. The results of the test launch of the system are assessed and a forecast is made about the possibility of the telecommunications operator to achieve its goals through the digital transformation of the loyalty system.
- Stage 5 – industrial exploitation. the system is launched into operation for the full network of customers. Technical and customer support is also provided. The difference lies only in the scope of the proposed improvements and in the possibility of improving the loyalty system only in terms of operational processes.
- Stage 6 – summing up the digital transformation. It is determined whether the goal of digital transformation has been achieved, as well as the problems faced by the company.

In this paper we proposed the methodology of the assessment of the loyalty system's digital transformation as a realization of Stage 4. And also, we show how to implement this methodology on the example of Albania's telecommunication operator.

The digital transformation of loyalty systems is a complex shift in customer relations in the company's business model, which entails the reengineering of the cultural and operational parts of the

Table 1: Description of costs in calculating the effectiveness of the digital transformation of the loyalty system

Cost type	Cost description
The cost of developing a digital transformation strategy	The cost of salaries for digital transformation managers The costs of expert analysis and evaluation of the digital maturity of the telecom operator The costs of evaluating infrastructure and business processes to assess the feasibility of digital transformation Road map development costs
The cost of developing a digital transformation loyalty program	The cost of salaries for marketers Competitor research costs Market research costs The costs of researching the sales and customers of the mobile operator
Costs of culture development and staff	Costs of hiring employees who will support the loyalty system Employee training costs
Loyalty system development costs	Payroll costs for the system development team Development software costs Hardware costs Hardware maintenance costs

Source: developed by the authors

company. Therefore, it is necessary to use an integrated approach for assessing the effectiveness of the digital transformation in this case.

In this regard, the assessment of the effectiveness of the loyalty system digital transformation is an assessment obtained as a result of the simultaneous study of a set of indicators reflecting strategic, financial, operational, cultural, technological aspects, and containing conclusions about the results of activities based on identifying differences from the comparison base. The comparison base is the performance indicator before the digital transformation of the loyalty system.

Thus, we have identified performance indicators (the ratio of growth in benefits and the costs of their implementation) and indicators of the effects of digital transformation of loyalty systems (group-strategy, client, operational, cultural, technological indicators).

In order to reveal the economic profit from the activity of the loyalty system, the following effects were taken into consideration: the customer growth numbers, the average purchase order of loyal offers, the partners' growth numbers, the average purchase check of the partner.

Profit growth is calculated according to the formula (1):

$$\Delta P = Q_c * P_c + \Delta Q_s * P_s * i \tag{1}$$

where Q_c is the number of customer growth over a certain period;

P_c – average receipt for purchases of loyal offers of the telecom operator for a certain period;

ΔQ_s – the number of partners' growth over a certain period;

P_s – the average receipt for partner purchases for a certain period;

i – the established percentage for partner transactions.

If the profit formula does not differ, regardless of the fact of the transformation carried out, then the digital transformation directly affects the calculation of costs.

The cost calculation formula after the digital transformation is calculated according to the formula (2):

$$C = C_{DTLS} + C_{op} \tag{2}$$

where C_{DTLS} is the cost of digital transformation;

$\mathrm{C_{op}}$ – the cost of maintaining the loyalty system for a certain period.

The full list of costs that are taken into account in calculating the efficiency of an enterprise after digital transformation is given in Table 1.

The calculation of the effectiveness of the loyalty system before digital transformation is calculated according to the classical formula (3):

$$E_{\mathrm{DTLS}} = \frac{\Delta P}{\mathrm{C_{op}}} \tag{3}$$

where ΔP is the increase in profit from the loyalty system for a certain period;

C_{op} – the cost of supporting the operation of the loyalty system for the period corresponding to the profit;

The calculation of efficiency after the transformation of the loyalty system is calculated according to the formula (4):

$$E_{DTLS} = \frac{\Delta P}{C_{DTLS} + C_{op}} \tag{4}$$

where C_{DTLS} is the cost of carrying out the digital transformation of the operation of the transformed loyalty system.

The calculation of economic efficiency is made on the basis of profit and cost indicators for the same period. Since digital transformation is a long-term process that requires large investments. The final economic efficiency of this measure is often achieved after 2 years, but each company separately chooses its own period for assessing economic efficiency.

If the obtained value of the coefficient is $E_{DTLS} \geq 1$, then this transformation was beneficial for the telecommunications operator company and the investment in it has already paid off for the calculated period.

If the obtained value of the coefficient is $E_{DTLS} < 1$, then this transformation did not pay off for the calculated period. But it is worth noting that the digital transformation of the loyalty system is an expensive undertaking, the payback of which must be considered in the long term, while pondering economic indicators in an inextricable connection with non-economic ones.

To assess the indicators of the effects of digital transformation of the loyalty system of a telecommunications company, we identified 5 areas (groups of indicators of strategic effect $k1$, customer effect $k2$, operational effect $k3$, effect in the company's personnel culture $k4$, effect in the use of technologies $k5$). The Table 2 contains the characteristics of these groups of indicators.

At the stage of summing up the results of digital transformation, target indicators that allow fixing the main vectors of progress are calculated. Typical indicators are categorized by major digital transformation vectors. With the help of this classification, companies will be able to identify the most successful areas of digital transformation, as well as areas in which more detailed study is required to achieve the target effect.

The determination of performance in each direction of the company's activities within the framework of digital transformation is carried out according to Table 3.

In order to get an "excellent" result in each of the areas of digital transformation, it is necessary that each indicator matches to the target indicator set in the strategic document by the telecommunications operator.

As noted earlier, the assessment of the effectiveness of the loyalty system digital transformation should study "a set of indicators reflecting the strategic, financial, operational, cultural, technological aspects". Table 4 presents a matrix for assessing the digital transformation of the loyalty system according to the proposed method.

The evaluation of the effectiveness of digital transformation occurs at two stages: the assessment of intermediate indicators, summing up the results of digital transformation. When evaluating intermediate indicators, the company identifies the effects of applying the developed solution. The company then compares the intermediate results with the target intermediate indicators and decides on the development measures.

3 RESULTS

The common threat to telecommunications brands is the small or absent personal contact with a client: as soon as a subscription to a service expires, customers who do not have a loyal relationship to the company are often interested in what a competitor offers. Under quarantine conditions, this problem manifested itself even more strongly. During the quarantine season in Russia, major telecommunications companies, such as VympelCom and MTS began to lose their subscriber base. But, despite the decline in their subscribers' base, MTS demonstrated a growth in revenue during 2020 due to an increase in highly profitable subscribers who consume more than one service from the telecommunications operator. At the beginning of the pandemic, the telecommunications company Vympelcom was unable to respond quickly to the changing market; therefore, in the second quarter of 2020, its number of subscribers decreased by 8% compared to the same period in 2019, and the company's revenue over the same period decreased by 10%.

To analyze the loyalty systems, the largest foreign telecommunications companies from America, Europe and Asia were selected. American multinational telecommunications conglomerate Verizon; the second largest telecommunications provider in the O2 UK (26 million subscribers as of November 2019); the American trade agency Telarus, which concludes contracts with commercial network and cloud providers; Thailand's largest mobile operator GSM (with over 40 million customers) and Albania's telecom operator.

As a result of the analysis, conclusions were made about the decrease in the possibility of scaling companies due to drop in the number of new sales markets; an increase in the load on information systems with the growth of the Internet of Things; the need to solve the problem of low customer loyalty in the post-covid market conditions.

At the same time, the main problems of loyalty programs of telecommunication companies were identified, including: the acquisition of unsuitable software (loyalty systems); untimely implementation of the loyalty system, complicated processes of registration, accrual or paying-off; insufficient interface development; insufficient capacity of the selected information system; loyalty programs that are not integrated into the mobile application lead to the necessity of carrying a loyalty card or printed coupons with the costumer; loyalty system fraud.

So, the Albania's telecom operator has been considered as the object for assessment of the digital transformation of the loyalty systems. The digital maturity of the company and the processes of building loyal relationships with customers were assessed. And analysis of such indicators as the nature of building relationships, the index of customer satisfaction with the program and the loyalty system, the use of technologies for the selection of customer-oriented offers, the method of access to the loyalty system and the amount of customer churn led to the conclusion about the need for digital transformation.

On the second stage a strategy for the digital transformation of an Albania's telecommunication company was prepared with orientation at the eTOM model.

Digital transformation of customer loyalty systems for telecommunications companies consists in the use of modern technologies to modernize business processes. In particular, it is done through the combination of the business processes management of a telecommunications company and management of communication networks, implemented in the systems of business support and operational activities: Operations Support System (OSS) and Business Support System (BSS). Network designers, service designers, support services, architects, operations and engineering teams, commonly use OSS. BSS includes software applications that typically support customer-centric activities. BSS is a frontal OSS application system. The OSS usually transmits various service orders, and also provides information on ensuring the BSS servicing. The main property of such global control systems is independence from the type of implemented software and the integration of processes approach. But there is a factor limiting the implementation of programs of this type: the lack of a unified standard for presenting data in the system, business processes and interfaces for interacting with the company's environment.

The International Telecommunication Union proposed to use documents developed by the Telemanagement Forum (TM Forum)

Table 2: Criteria for evaluating the effects of digital transformation in a telecommunications company

Group of indicators	Indicator	Definition
Indicators of strategic effect, $k1$	Success of customer self-service	the amount of money that the customer is expected to spend during his entire relationship with the organization, minus the costs of their acquisition and maintenance
	Return on investment (ROI)	indicator of profitability of investments
	Digital Investment Income (RDI)	additional income received as a result of digital investments
	Brand commitment indicator	indicator of the degree of consumer support for a particular brand
	Revenue received by partners	the share of the company's revenue received from services jointly provided with one or more partners under the loyalty system
Customer impact indicators, $k2$	Lifetime value of the customer	the amount of money that the customer is expected to spend with the organization throughout their relationship, minus the costs of their acquisition and maintenance
	The Consumer Loyalty Index (NPS)	an indirect indicator of customer satisfaction and loyalty based on the likelihood that customers will want to recommend a loyalty program
	Duration of the system response time	the time (duration) required for the operator to provide an initial meaningful response to the client's request
	Customer churn indicator	customer turnover indicator of the loyalty program
	The cost of attracting customers	the average cost of an organization to attract one customer to a loyalty program
Operational effect indicators, $k3$	Time to market the offer (TTM)	the time spent on the transition of a new or improved loyalty system offer from concept creation to market launch
	Average response time	the time required to provide the results of a request made by a customer in the loyalty system
	Average usage level	the number of users from the user base who actively interact with the loyalty system. This is an indicator show how effective the loyalty program and system are
	The cost of service	an indicator of the total cost of customer service of the loyalty program.
Indicators of the effect in the company's personnel culture, $k4$	Internal Consumer Loyalty Index (INPS)	an indicator that allows you to assess the degree of readiness of employees to recommend a loyalty program
	Compliance of personnel skills with digital transformation	an indicator of the availability of personnel with the skills necessary for the implementation of digital initiatives
	Employee effort indicator	an assessment of the degree of complexity that an employee faces when developing and/or supporting digital products and services of a company
	The indicator of personnel involvement in digital transformation	the working time of employees engaged in the development, delivery and support of a loyalty system, expressed as a percentage of total working time
	The share of funds allocated for staff training	the training budget for digital initiatives measures the percentage of the allocated budget for training digital initiatives from the share of the total budget allocated to the needs of the company
Indicators of the effect in the use of technologies, $k5$	Indicator of processes automation	the proportion of system processes that can be fully implemented through digital channels without interaction with a person (that is, without the support of company personnel)
	Rating of the loyalty system application	quality and usability of the application. It is determined based on such indicators as the star rating of the application, the total number of downloads/ installations, the total number of ratings and the average mood of users
	DevOps Implementation	a measure indicating the level of DevOps implementation (CI/CD) in technology delivery operations within the loyalty system
	The level of client orientation of the application interface	determined by several indicators: the speed of finding the necessary control, the percentage of completed operations, the level of application usage by type of functionality

Source: developed by the authors

Table 3: Determination of performance in the direction of digital transformation

The direction of digital transformation					The result of the effect in the direction
$k1 > 1$	$k2 > 1$	$k3 > 1$	$k4 > 1$	$k5 > 1$	Excellent
$k1 < 1$	$k2 > 1$	$k3 > 1$	$k4 > 1$	$k5 > 1$	Good
$k1 < 1$	$k2 < 1$	$k3 > 1$	$k4 > 1$	$k5 > 1$	Satisfactorily
$k1 < 1$	$k2 < 1$	$k3 < 1$	$k4 > 1$	$k5 > 1$	Unsatisfactory

Source: developed by the authors

Table 4: Matrix for evaluating the digital transformation of the loyalty system according to the proposed method

Indicators of the effectiveness of digital transformation	Groups of indicators of the effects of digital transformation
Efficiency coefficient before Digital Transformation	Strategic
Efficiency coefficient after digital transformation	Customer oriented
Comparison of performance indicators	Operational
	Personnel culture in the company
	Information processing and storage

Source: developed by the authors

as industry standards. For example, they propose to use an extended telecom operator Business Process Framework (eTOM) as a structural model of the business processes of a telecommunications company. eTOM is a reference architecture that takes into account the business processes possible in a telecommunications company. There is a decomposition of the levels of development of business processes, where levels 1–3 are standardized in the specifications of the eTOM architecture [27], these are at the top levels and are independent of the characteristics of the company and the services it provides. Level 4-6 processes are unique to each company. The eTOM model implies more levels of decomposition, but the levels below the third are technology-dependent and are already determined during the construction of real processes and operational support systems.

Authors use this model as base to design a hybrid architecture for dynamic service delivery with improved resource performance and quality of service for telecommunication companies. For example, Seraoui Y. et all [28] suggested a mapping of this eTOM business process model upon the network functions virtualization (NFV) framework for mobile network operators using 5G technology. Decomposition of the “Order acceptance and processing” scenario of the second level of the eTOM model presented at figure 1.

The revision of the eTOM model is to replenishment of the Marketing Sales Domain, the Domain of service operations and the Client domain taking into account the functioning of loyalty systems (highlighted in purple). The "Order Acceptance and Processing" script manages the process of preparing and completing the order, starting with the feasibility check and ending with the activation of services and resources. The scope of the loyalty system in this scenario extends to the domains of the client and marketing sales.

The Marketing Sales Domain in the Accomplishment sector is proposed to add the Identifying potential customers. When there are receiving and processing an order the loyalty system must identify potential customers for each service in the catalog. Client domain should be modified the Operational process support, the Accomplishment and the Quality assurance at the same time. Because of the loyalty system performs such functions as selection of customer-oriented services based on customer data; recommendation of loyal offers; accrual and deduction of loyalty points for an order; compensation for poorly and/or untimely service rendered. The added functions of the loyalty system are interrelated. The error in performing one of the functions may lead to incorrect operation of the loyalty system as a whole. The last added process related to the loyalty system is to Operational process support. This process is aimed at detecting fraud in real time.

So, the strategy of loyalty system’s digital transformation of Albania’s company had included several connected directions and technologies implementation based on the eTOM model modification:

- using machine learning to create a predictable model based on collaborative filtering as a component of the selection of personalized bonus offers;
- using OSS/BSS approach to software type in an architecture of the loyalty system;
- using the Big Data in data storage by the RAW storage ("Data Lake" method) to providing the access to complete primary data;
- using Data Leak Prevention (DLP) technology as a tool for monitoring the daily work of employees. It allows company to find security weaknesses before an incident occurs.

In 2018-2019 at the Albania’s telecommunications company the new technologies were introduced into business processes, including the introduction of a customer loyalty system. As a Stage 3 realization a predictive analytics component, a personalized bonus offers matching component, an OSS / BSS program architecture, and a security component were implemented.

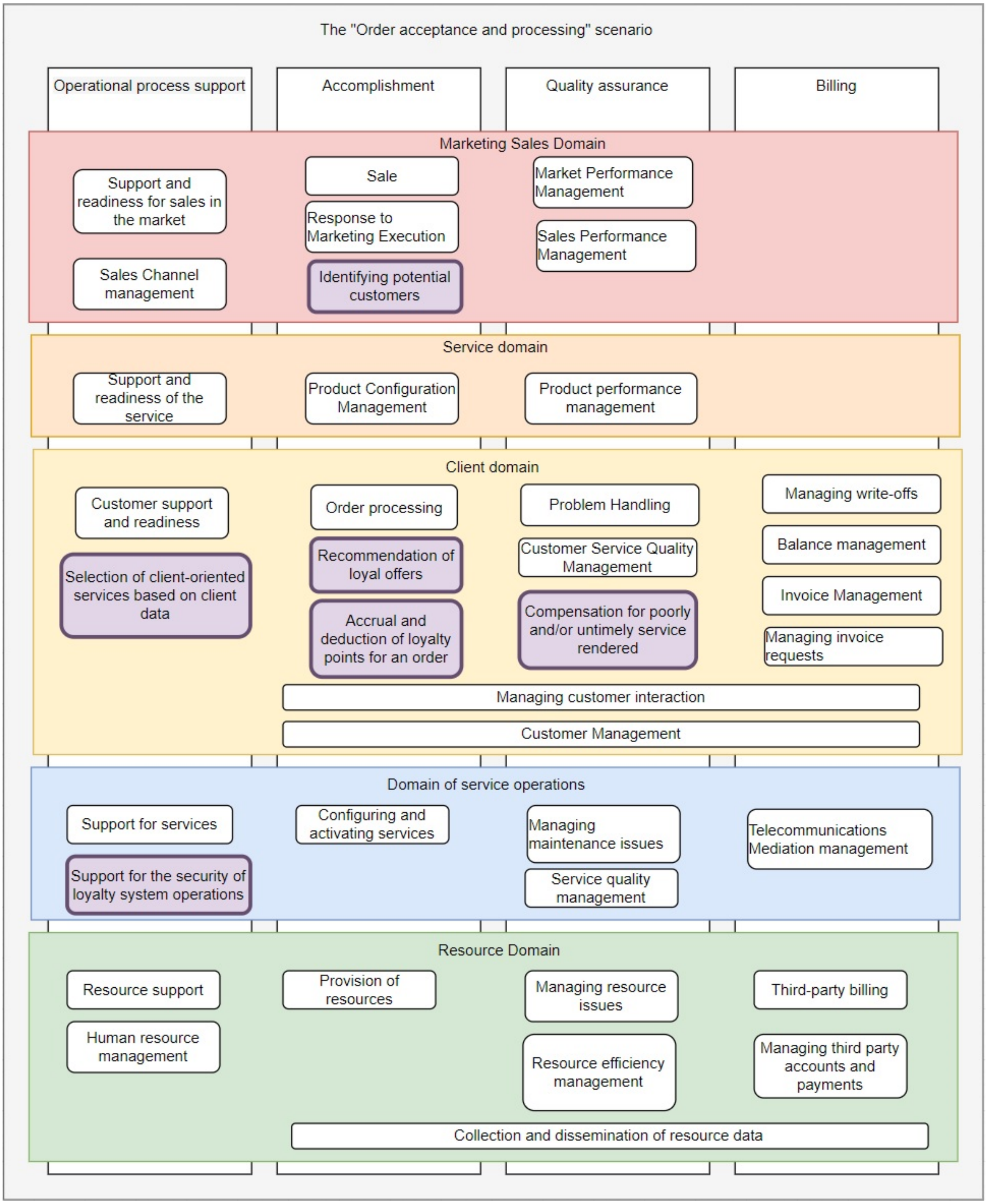

Figure 1: Decomposition of the "Order acceptance and processing" scenario of the second level of the eTOM model [29, 30]

The assessment of intermediate indicators of the effectiveness of loyalty system's digital transformation we need to start with the calculation its economic efficiency (Table 5).

The efficiency ratio of the loyalty system with a value less than 1 allows us to conclude that, before the digital transformation, the loyalty system did not compensate the expenses and the existing system did not meet the needs of the business in the telecommunications sector. After the digital transformation, the value of the loyalty system efficiency indicator greater than 1 means that, over the calculated period of time, the company has built effective loyal

Table 5: Results of the economic efficiency of the loyalty system before and after digital transformation

The name of the indicator	Before transformation	After transformation
Increment for 1 year, euro	96 348	196 678
Increment for 2 year, euro	78 905	245 993
Loyalty system support 1-st year, euro	124 784	
Loyalty system support 2-nd year, euro	157 456	
Digital transformation costs, euro		334 671
Digital transformation support for 1-st year, euro		38 554
Digital transformation support for 2-nd year, euro		33 456
Efficiency coefficient	0,62	1,08

Source: developed by the authors based on [31]

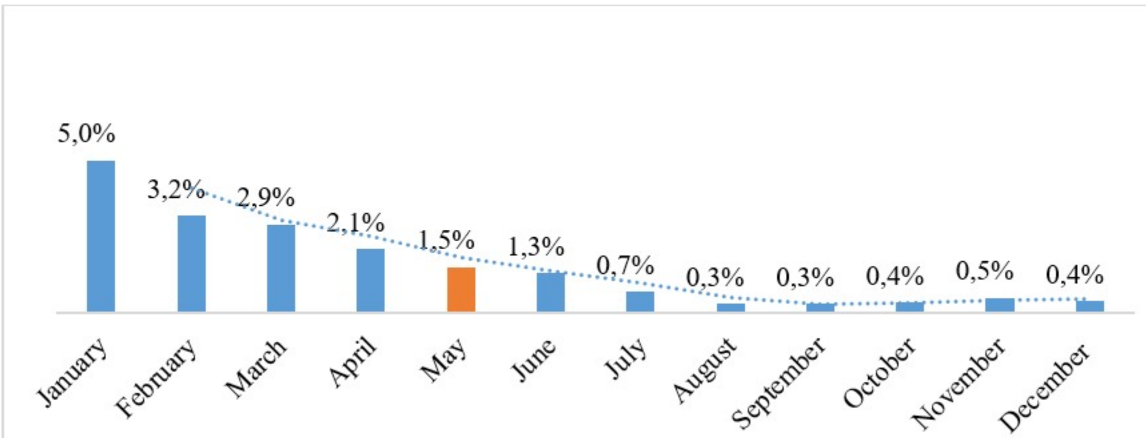

Figure 2: Dynamics of customer outflow of Albania's telecom operator in 2019 [31]

relationships with customers, which indirectly bring profit to the company as a whole.

Further, the indicators of the effects of digital transformation of loyalty systems that are available were calculated. And the most obvious indicator of the effectiveness of the implementation of the loyalty system is the growth in the number of participants in the loyalty system. After the introduction of the Vodafone Albania operator loyalty system, the number of members has grown from 50% in January 2019 to 92.3% in December 2019 (Figure 2). At the same time, there is a decrease in customers leaving from 5.0% in January 2019 to 0.4% in December 2019 (as a percentage of the total customers leaving of the Vodafone Albania telecom operator).

The second most important indicator of the effectiveness of the loyalty program is the aggregate of customer loyalty indicators, reflecting customer satisfaction with the company's performance (indicators and indices of customer satisfaction). The assumption is that the more satisfied a customer is, the more likely they are to keep being a customer. The assessment of the customer satisfaction index (CSI) is often based on the following parameters: cost, quality, assortment, staff work, service, reliability and others. Often the CSI index is measured together with the NPS (Net Promoter Score) loyalty index, creating a customer journey map, evaluating the quality of services provided using the Mystery Shopping method.

In this study, the CSI was assessed before and after the digital transformation of the loyalty system at Albania's telecommunication company. According to the results of the study, the CSI index was raised from 51% to 79% (Table 5), which indicates a high level of customer satisfaction with the service provided.

The next indicators of the effectiveness of digital transformation of loyalty programs were defined. These indicators partially coincide with those proposed by the authors, but the company has identified only 9 such indicators. Albania's telecommunication company non-cost-effective loyalty system before and after digital transformation, according to the indicators available for research, are shown in Table 6.

As the reader can see, parameters that are important for customers with below average satisfaction levels were not discovered. Indicators of the effects of digital transformation of the loyalty system show that digital transformation has been successful for the Albania's telecommunications operator.

An unsatisfactory effect in the variable of personnel culture was created due to the low indicators of the internal index of consumer loyalty (INPS), the correspondence of the skills of the digital transformation personnel, and the indicator of the personnel's involvement in digital transformation. As a result of this assessment,

Table 6: Indicators of the effects of the digital transformation of the Albania's operator loyalty system

Indicator name	Before transformation	Target value	After transformation
Success of customer self-service, %	16,8%	25%	38%
The Consumer Loyalty Index (NPS), %	51%	75%	79%
Customer churn indicator, %	37,5%	27%	25%
The average client connection time, hours	3-4	20	from 15 minutes to 1,5 hours
Time to market loyalty offers (TTM), hours	16-24	10	4-8
Internal Consumer Loyalty Index (INPS), %	34%	75%	57%
Indicator of processes automation, %	53%	70%	73%
Share of funds allocated for staff training, %	3%	7%	5%
Compliance of personnel skills with digital transformation	low, requires to be developing	high	Average, still requires to be developing

Source: data from the Report on the transformation of the loyalty system of Albania's telecommunication company, compiled with the participation of the authors

Table 7: Evaluation of the digital transformation of the Albania's operator loyalty system

Performance indicators	Group of indicators of digital transformation effects
Efficiency coefficient before digital transformation, E < 1	Strategic, $k1 > 1$
Efficiency coefficient after digital transformation, EDTLS > 1	Customer oriented, $k2 > 1$
Comparison of performance indicators	Operational, $k3 > 1$
	Personnel culture in the company, $k4 < 1$
	Information processing and storage, $k5 > 1$

Source: developed by the authors

we conclude that the company has to adjusted its digital transformation activities to go on Stages 5 and 6. The company need to conclude agreements as soon as possible to train personnel in new technologies, and a large-scale immersion of the company's personnel in digital transformation was carried out.

Thus, by using the suggested method, a matrix for evaluating the digital transformation of the loyalty system of Albania's telecommunication company was created (Table 7).

As a result, it can be concluded that the digital transformation of the Albania's telecommunication operator loyalty system has been successful. Target indicators were achieved in most areas of digital transformation, with the exception of the area of "Personnel culture in the company". At the intermediate assessment stage of digital transformation, this area was located in the "red" performance zone. Albania's telecommunication company was able to increase the effectiveness of this area to a satisfactory result, but still its activity in this direction was not fully successful.

For this reason, we can conclude that even while having only a satisfactory result in the area of "Personnel culture in the company", Albania's telecommunication company has put into operation effective mechanisms for building loyal relationships with the customers.

4 DISCUSSION

The study proved that the digital transformation management in telecommunication companies based on eTOM modified model and on complex approach has a positive effect on the company's efficiency. We identified and evaluated the positive effects of the technologies implementation into the loyalty systems of the telecommunication companies as the basis of their competitiveness in view of current trends in reducing user growth.

At the same time using technologies might be expensive and lead company to loses. So digital transformation management should pass all 6 stages, including such instruments as the digital roadmaps [32] or Industry 4.0 Maturity Index approach [33], as well as the forecasting models of customer retention using various methods [26].

The existing studies developed a methodology to manage digital transformation for SMEs, and our work is devoted to the transformation of telecommunication companies in particular, the loyalty system. Our research results present the most effective measures for a successful loyalty system's digital transformation of telecommunication company as the one of core component of its business processes and its effectiveness.

5 CONCLUSION

The paper found that pace of changes and developments in the telecommunications market are taking on a large-scale character. Optimization of business processes of a telecommunications company is due to the high knowledge intensity of the industry, the peculiarity of services, namely the short life cycle and continuity of their provision. The most of the proposals for optimizing business processes are aimed at automating the process, i.e., the introduction of information technology. Although the use of modern technologies is one of the decisive factors of successful transformation, but the success of the transformation of the loyalty system consists in the totality of the efforts made. The institutional and the operational parts of the telecom operator's company business processes should be adapted from the point of view of new methods of building loyal relationships with the client.

We have proposed the methodology of the assessment the of customer loyalty system digital transformation effectiveness as a part of management approach to digital transformation of telecommunication company. The assessing of the effectiveness of the customer loyalty system's digital transformation Albania's telecommunication company with two blocks of indicators: economic efficiency coefficients and a group of indicators of effects accompanying. So, it shows an increase in the efficiency coefficient from 0.62 to 1.08, as well as an increase in all groups of effects except of personnel culture in the company. In particular, the customer satisfaction index (CSI) increases from 51% to 79% and the outflow of customers decrease from 37.5% to 25%. Finally, there was a discussion of the results of our study, and the conclusions were summarized.

The next study directions will be connected with identifications key effectiveness indicators of a company's digital transformation and factors influencing on digital transformation processes.

The features and trends in the development of loyalty programs in telecommunications companies were identified, criteria for assessing the effectiveness of digital transformation of loyalty programs were considered, the effectiveness of digital transformation of loyalty programs of a telecommunications company was assessed, and the study proved that the introduction of technologies has a positive effect on the effectiveness of company in the telecommunications sector.

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